

DAX & Data Visualization

DAX Calculations

Calculated Columns

- Create a Calculated Column for 'Category Type':

1 Category Type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]							
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	
B-25628	24	-9	4	Clothing	Saree	Clothing - Saree	
B-25633	711	-8	4	Clothing	Saree	Clothing - Saree	
B-25635	382	30	3	Clothing	Saree	Clothing - Saree	
B-25636	637	113	5	Clothing	Saree	Clothing - Saree	
B-25640	122	-47	4	Clothing	Saree	Clothing - Saree	
B-25646	20	-8	2	Clothing	Saree	Clothing - Saree	
B-25647	42	-6	4	Clothing	Saree	Clothing - Saree	
B-25648	55	-26	4	Clothing	Saree	Clothing - Saree	
B-25648	130	-41	4	Clothing	Saree	Clothing - Saree	
B-25650	211	-105	2	Clothing	Saree	Clothing - Saree	
B-25650	31	-2	2	Clothing	Saree	Clothing - Saree	
B-25650	512	-225	5	Clothing	Saree	Clothing - Saree	
B-25650	238	20	2	Clothing	Saree	Clothing - Saree	
B-25651	23	4	1	Clothing	Saree	Clothing - Saree	

- Calculate Revenue per Order in Order Details Table:

✕ ✓		1 Revenue Per Order = 'Order Details'[Amount] * 'Order Details'[Quantity]					
		2					
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue Per Order
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	1683
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	952
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	579
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	1413
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	525
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	100
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	129
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	320
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	14427
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	2824
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	1602
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	596
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	3175
B-25628	24	-9	4	Clothing	Saree	Clothing - Saree	96
B-25633	711	-8	4	Clothing	Saree	Clothing - Saree	2844
B-25635	382	30	3	Clothing	Saree	Clothing - Saree	1146
B-25636	637	113	5	Clothing	Saree	Clothing - Saree	3185
B-25640	122	-47	4	Clothing	Saree	Clothing - Saree	488
B-25646	20	-8	2	Clothing	Saree	Clothing - Saree	40
B-25647	42	-6	4	Clothing	Saree	Clothing - Saree	168
B-25648	55	-26	4	Clothing	Saree	Clothing - Saree	220
B-25648	130	-41	4	Clothing	Saree	Clothing - Saree	520
B-25650	211	-105	2	Clothing	Saree	Clothing - Saree	422
B-25650	31	-2	2	Clothing	Saree	Clothing - Saree	62
B-25650	512	-225	5	Clothing	Saree	Clothing - Saree	2560
B-25650	238	20	2	Clothing	Saree	Clothing - Saree	476
B-25651	22	4	1	Clothing	Saree	Clothing - Saree	22

- Create a Calculated Column to Categorize Sales:

✕ ✓ 1 Sales Category = 2 IF(3 'Order Details'[Amount] > AVERAGE('Order Details'[Amount]), 4 "Above Average", 5 "Below Average"								
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue Per Order	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	1683	Above Average
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	952	Below Average
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	1413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	525	Below Average
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	14427	Above Average
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	2824	Above Average
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	1602	Above Average
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	596	Below Average
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	3175	Above Average
B-25628	24	-9	4	Clothing	Saree	Clothing - Saree	96	Below Average
B-25633	711	-8	4	Clothing	Saree	Clothing - Saree	2844	Above Average
B-25635	382	30	3	Clothing	Saree	Clothing - Saree	1146	Above Average
B-25636	637	113	5	Clothing	Saree	Clothing - Saree	3185	Above Average
B-25640	122	-47	4	Clothing	Saree	Clothing - Saree	488	Below Average
B-25646	20	-8	2	Clothing	Saree	Clothing - Saree	40	Below Average
B-25647	42	-6	4	Clothing	Saree	Clothing - Saree	168	Below Average
B-25648	55	-26	4	Clothing	Saree	Clothing - Saree	220	Below Average
B-25648	130	-41	4	Clothing	Saree	Clothing - Saree	520	Below Average
B-25650	211	-105	2	Clothing	Saree	Clothing - Saree	422	Below Average

Calculated Measures

1. Order Count:

Order Count = COUNTROWS('Order Details')								
Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue Per Order	Sales Category	Order Count
561	212	3	Clothing	Saree	Clothing - Saree	1683	Above Average	1500
119	-5	8	Clothing	Saree	Clothing - Saree	952	Below Average	1500
193	-166	3	Clothing	Saree	Clothing - Saree	579	Below Average	1500
157	5	9	Clothing	Saree	Clothing - Saree	1413	Below Average	1500
75	0	7	Clothing	Saree	Clothing - Saree	525	Below Average	1500
25	-5	4	Clothing	Saree	Clothing - Saree	100	Below Average	1500
43	0	3	Clothing	Saree	Clothing - Saree	129	Below Average	1500
160	-59	2	Clothing	Saree	Clothing - Saree	320	Below Average	1500
1603	0	9	Clothing	Saree	Clothing - Saree	14427	Above Average	1500
353	90	8	Clothing	Saree	Clothing - Saree	2824	Above Average	1500
534	0	3	Clothing	Saree	Clothing - Saree	1602	Above Average	1500
149	-87	4	Clothing	Saree	Clothing - Saree	596	Below Average	1500
635	-349	5	Clothing	Saree	Clothing - Saree	3175	Above Average	1500
24	-9	4	Clothing	Saree	Clothing - Saree	96	Below Average	1500
711	-8	4	Clothing	Saree	Clothing - Saree	2844	Above Average	1500
382	30	3	Clothing	Saree	Clothing - Saree	1146	Above Average	1500
637	113	5	Clothing	Saree	Clothing - Saree	3185	Above Average	1500
122	-47	4	Clothing	Saree	Clothing - Saree	488	Below Average	1500
20	-8	2	Clothing	Saree	Clothing - Saree	40	Below Average	1500
42	-6	4	Clothing	Saree	Clothing - Saree	168	Below Average	1500
55	-26	4	Clothing	Saree	Clothing - Saree	220	Below Average	1500
130	-41	4	Clothing	Saree	Clothing - Saree	520	Below Average	1500
211	-105	2	Clothing	Saree	Clothing - Saree	422	Below Average	1500
31	-2	2	Clothing	Saree	Clothing - Saree	62	Below Average	1500
512	-225	5	Clothing	Saree	Clothing - Saree	2560	Above Average	1500
238	20	2	Clothing	Saree	Clothing - Saree	476	Below Average	1500
23	4	1	Clothing	Saree	Clothing - Saree	23	Below Average	1500

2. Average Profit in Delhi:

1 Average Profit Delhi = 2 CALCULATE(3 AVERAGE('Order Details'[Profit]), 4 'List of Orders'[City] = "Delhi" 5)					
Order ID	Order Date	CustomerName	State	City	Average Profit Delhi
B-26002	08 February 2019	Harsh	Delhi	Delhi	25.6
B-26001	07 February 2019	Patil	Delhi	Delhi	7
B-26000	06 February 2019	Shubham	Delhi	Delhi	84.33333333333333
B-25999	05 February 2019	Diwakar	Delhi	Delhi	38.1818181818182
B-25996	04 February 2019	Prashant	Delhi	Delhi	39.2857142857143
B-25995	04 February 2019	Yohann	Delhi	Delhi	342
B-25994	03 February 2019	Omkar	Delhi	Delhi	-7
B-25993	03 February 2019	Harshal	Delhi	Delhi	123.428571428571
B-25990	02 February 2019	Mugdha	Delhi	Delhi	32
B-25989	01 February 2019	Anjali	Delhi	Delhi	31.2
B-25988	31 January 2019	Nirja	Delhi	Delhi	24
B-25987	31 January 2019	Manjiri	Delhi	Delhi	5.5
B-25984	30 January 2019	Kajal	Delhi	Delhi	97
B-25983	30 January 2019	Atul	Delhi	Delhi	-24.4
B-25982	29 January 2019	Hemangi	Delhi	Delhi	3
B-25981	28 January 2019	Amruta	Delhi	Delhi	62.83333333333333
B-25914	18 December 2018	Kajal	Delhi	Delhi	82.5
B-25912	16 December 2018	Preksha	Delhi	Delhi	11
B-25910	14 December 2018	Jay	Delhi	Delhi	-44.16666666666667
B-25908	12 December 2018	Pradeep	Delhi	Delhi	25
B-25906	11 December 2018	Abhijit	Delhi	Delhi	27
B-25904	10 December 2018	Swapnil	Delhi	Delhi	71.5
B-25913	17 December 2018	Geetanjali	Madhya Pradesh	Delhi	21
B-25909	13 December 2018	Sujay	Madhya Pradesh	Delhi	90

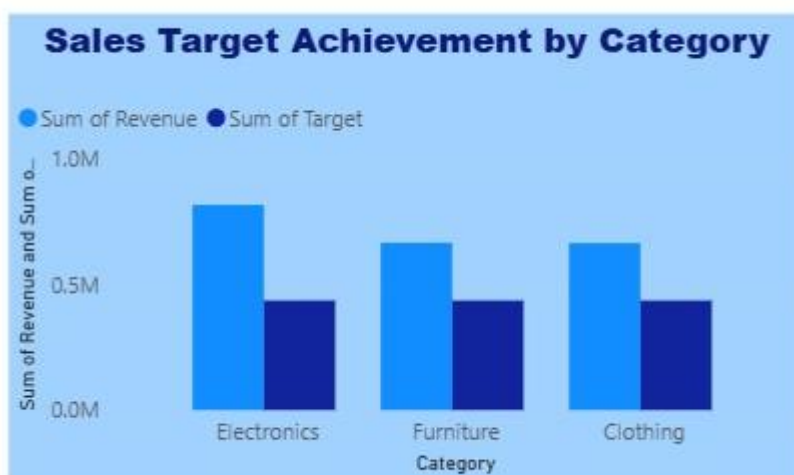
3. Year-to-Date (YTD) Sales:

YTD Sales = TOTALYTD(SUM('Order Details'[Revenue]), 'List of Orders'[Order Date]) 											
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category	Order Count	YTD Sales	Pro
25602	561	212	3	Clothing	Saree	Clothing - Saree	1683	Above Average	1500	1683	0.01
25602	119	-5	8	Clothing	Saree	Clothing - Saree	952	Below Average	1500	952	0.01
25603	193	-166	3	Clothing	Saree	Clothing - Saree	579	Below Average	1500	579	0.01
25604	157	5	9	Clothing	Saree	Clothing - Saree	1413	Below Average	1500	1413	0.01
25605	75	0	7	Clothing	Saree	Clothing - Saree	525	Below Average	1500	525	0.01
25609	25	-5	4	Clothing	Saree	Clothing - Saree	100	Below Average	1500	100	0.01
25610	43	0	3	Clothing	Saree	Clothing - Saree	129	Below Average	1500	129	0.01
25611	160	-59	2	Clothing	Saree	Clothing - Saree	320	Below Average	1500	320	0.01
25613	1603	0	9	Clothing	Saree	Clothing - Saree	14427	Above Average	1500	14427	0.01
25619	353	90	8	Clothing	Saree	Clothing - Saree	2824	Above Average	1500	2824	0.01
25622	534	0	3	Clothing	Saree	Clothing - Saree	1602	Above Average	1500	1602	0.01
25623	149	-87	4	Clothing	Saree	Clothing - Saree	596	Below Average	1500	596	0.01
25625	635	-349	5	Clothing	Saree	Clothing - Saree	3175	Above Average	1500	3175	0.01
25628	24	-9	4	Clothing	Saree	Clothing - Saree	96	Below Average	1500	96	0.01
25633	711	-8	4	Clothing	Saree	Clothing - Saree	2844	Above Average	1500	2844	0.01
25635	382	30	3	Clothing	Saree	Clothing - Saree	1146	Above Average	1500	1146	0.01
25636	637	113	5	Clothing	Saree	Clothing - Saree	3185	Above Average	1500	3185	0.01
25640	122	-47	4	Clothing	Saree	Clothing - Saree	488	Below Average	1500	488	0.01
25646	20	-8	2	Clothing	Saree	Clothing - Saree	40	Below Average	1500	40	0.01
25647	42	-6	4	Clothing	Saree	Clothing - Saree	168	Below Average	1500	168	0.01
25648	55	-26	4	Clothing	Saree	Clothing - Saree	220	Below Average	1500	220	0.01
25648	130	-41	4	Clothing	Saree	Clothing - Saree	520	Below Average	1500	520	0.01
25650	211	-105	2	Clothing	Saree	Clothing - Saree	422	Below Average	1500	422	0.01

Data Visualization

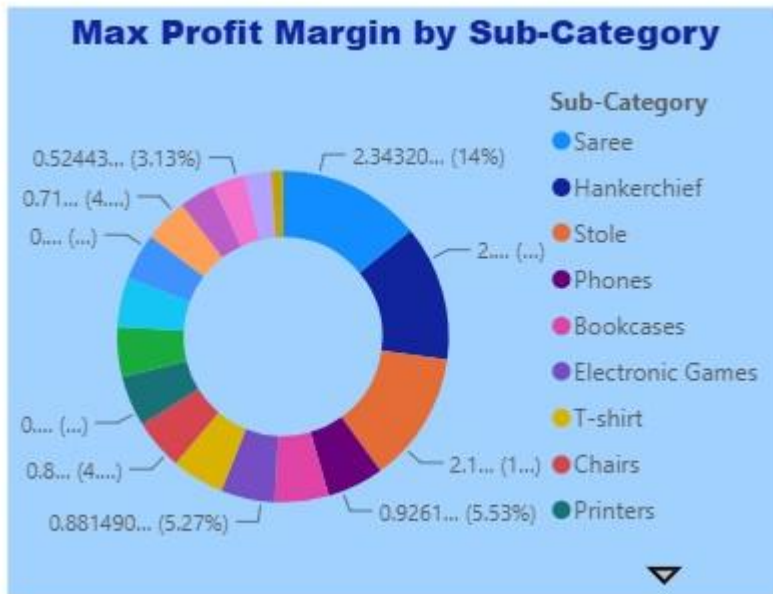
Create the following visualizations using Power BI:

- Sales Target Achievement by Category:



Comparing actual sales with sales targets by category using a clustered column chart.

- Max Profit Margin by Sub-Category:



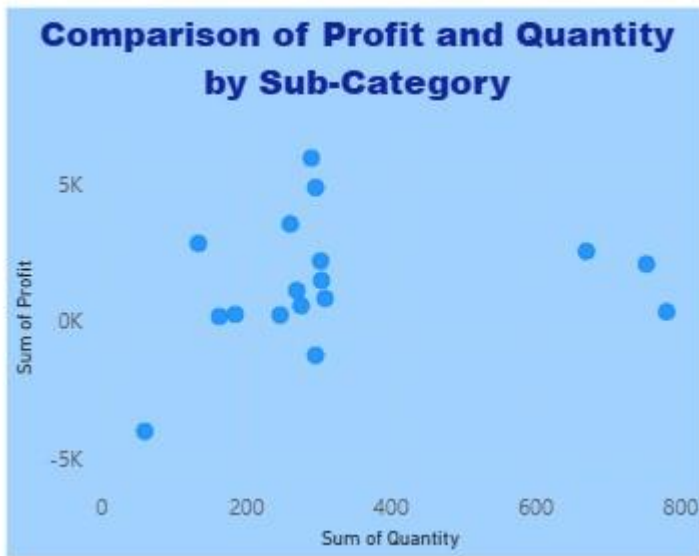
Analyzing the maximum profit margin for each sub-category of products using a donut chart.

- **Monthly Sales Trend:**



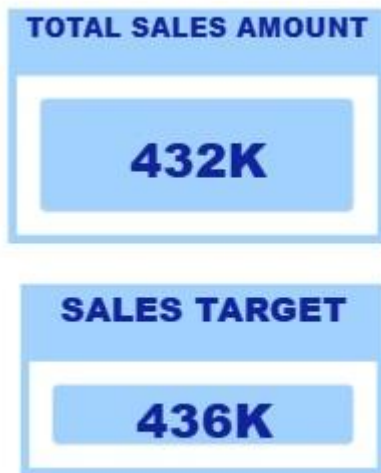
It Shows the trend of monthly sales over time using a line chart.

- **Comparison of Profit and Quantity by Sub-Category:**



Comparing the relationship between profit and quantity sold for different sub-categories using a scatter chart.

- Comparison of Total Sales Amount and Target:



I Created cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis.

I dont have a multi-row card to display the minimum target for each segment.

- Sales Performance Matrix:

Sales Performance Matrix						
MonthName	April			August		
Category	Sum of Actual Sales	Sum of Target Sales	Sum of Variance %	Sum of Actual Sales	Sum of Target Sales	Sum of Va
Furniture	733553400%	7410300	-17.15%	906154200%	9153900	
Electronics	949304400%	9589800	-22.20%	1035604800%	10461600	
Clothing	2675312400%	27025800	-62.55%	3236265000%	32692500	
Total	4358170200%	44025900	-101.90%	5178024000%	52308000	-

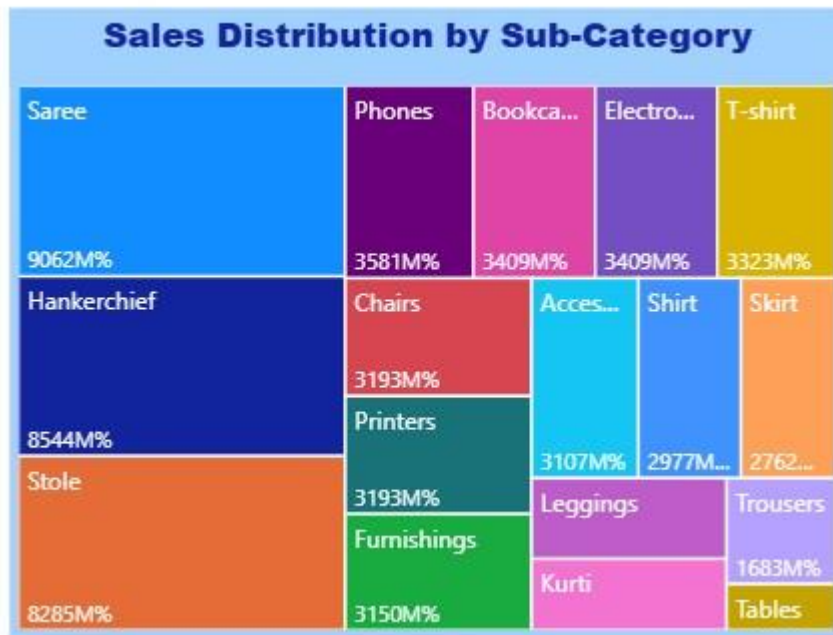
I Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.

- Geographic Sales Analysis:



It Visualize total sales on a map by city to identify regional sales patterns.

- Sales Distribution by Sub-Category:



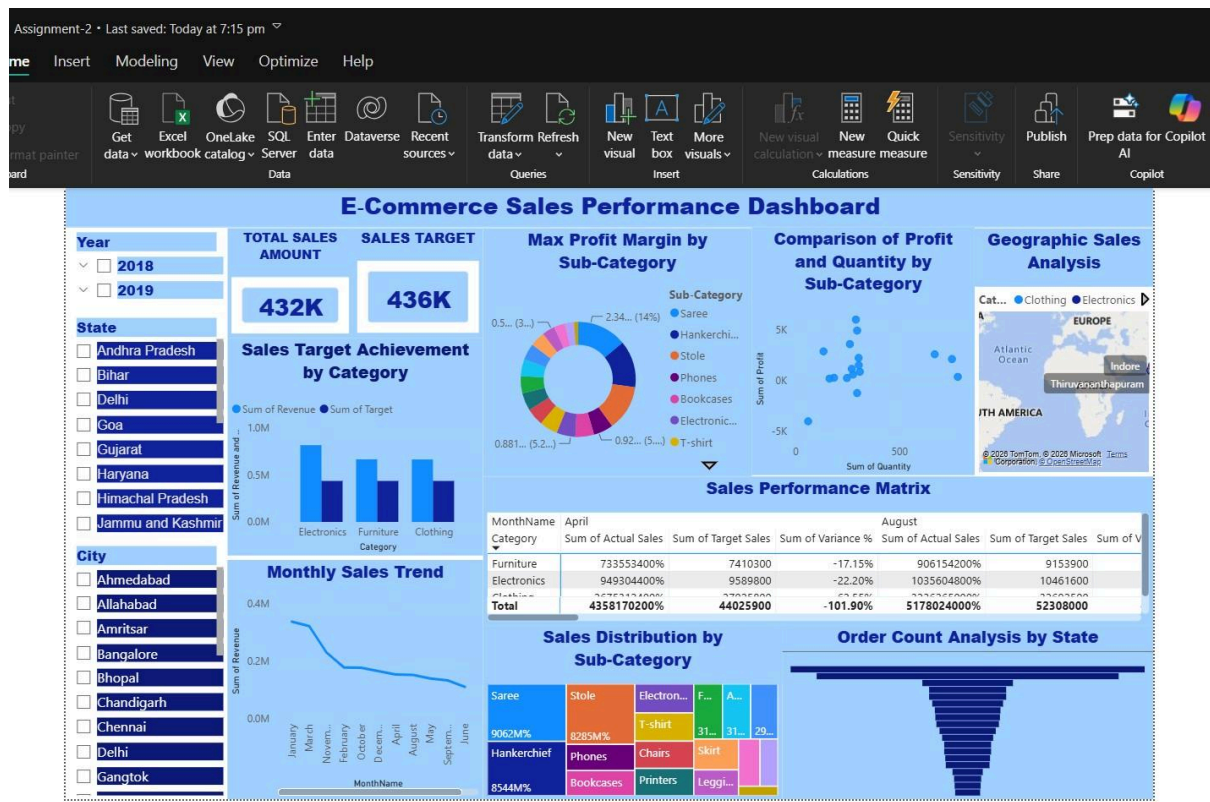
It Represent the sales distribution across different sub-categories using a treemap.

- Order Count Analysis by State:



I Created a funnel chart to visualize the distribution of order counts across different states.

- Dashboard layout:



Brief explanation for each visual:

Bar/Column Chart:

- Compare values across categories. Great for showing sales by region, product, or salesperson.

Line Chart:

- Display trends over time. Useful for tracking monthly revenue, growth rates, or customer acquisition.

Pie/Donut Chart:

- Show proportions of a whole. Best for illustrating market share or contribution of each product to total sales.

Scatter Chart:

- Plot relationships between two measures. Helpful for analyzing correlation (e.g., sales vs. marketing spend).

Matrix:

- Like a pivot table. Allows grouping by rows and columns with multiple measures (e.g., sales by region and product).

Card:

- Display a single KPI (e.g., Total Sales). Perfect for dashboards.

Map:

- Visualize geographic data. Useful for regional sales performance or customer distribution.

Treemap:

- Show hierarchical data as nested rectangles. Good for product category breakdowns.

Funnel Chart:

- Show stages in a process (e.g., sales pipeline). Highlights drop-offs between stages.

Slicer:

- Interactive filter for dashboards. Lets users filter by Year, city or State.